Tutorial sheet (07/05/2026)

Semiconductor Physics

1. Find the drift velocity of free electrons in a copper wire of cross-sectional area 10 mm^3 when the wire carries a current of 100 A. Assume that each copper atom contributes one electron to the free electron gas. Density of copper is 8969 kg/m^3 and its atomic weight is 63.54.**(7.4×10^{-4} m/s)**
2. Find the resistivity of intrinsic germanium with carrier density 2.5×10^{19} $/m^3$ at 300K. Given the mobilities of electron and hole as 0.39 and 0.19 m^2/Vs. **(0.43 ohm m)**
3. An N-type semiconductor has a resistivity of $20 \times 10^{-4}\Omega m$. The mobility of the electrons through a separate experiment was found to be 100×10^{-4} $m^2/Vsec$. Find the number of electrons per m^3. **($3.1 \times 10^{23}/m^3$)**
4. In an npn transistor circuit, the collector current is 15 mA. If 95% of the electrons emitted by the emitter reach the collector, what is the base current? (**0.78 mA**)
5. A transistor is connected in common-base configuration. If the emitter current is 2 mA and base current is 20 μA, what are the values of IC and α? **(1.98mA & 0.99)**

Optical fiber

1. An optical fiber has a core material of refractive index of 1.55 and cladding material of refractive index of 1.50. The light is launched into it in air. Calculate its numerical aperture. **(0.391)**
2. Calculate the angle of acceptance of a given optical fiber, if the refractive indices of the core and the cladding are 1.563 and 1.498 respectively. **(26°30')**
3. The numerical aperture of an optical fiber is 0.39. If the difference in the refractive indices of the material of its core and the cladding is 0.05, calculate the refractive index of the core material. **(1.2334)**
4. Calculate the fractional index change for a given optical fiber if the refractive indices of the core and cladding are 1.563 and 1.498 respectively. **(0.0416)**
5. Calculate the numerical aperture and the acceptance angle for an optical fiber with core and cladding refractive indices being 1.48 and 1.45 respectively. **[NA = 0.2965, θ_I (max) = 17°15']**
6. A glass clad fibre is made with core glass of RI 1.5 and cladding is doped to give fractional index difference of 0.0005. Find: a. cladding index and b. Acceptance angle **(1.499 and 0.17°)**
7. What is the attenuation in dB/km, if 15% of the power fed at the launching end of the ½ km fibre is lost during propagation? **(1.41 dB/km)**
8. A step index fibre is made with core of refractive index 1.52, a diameter of 29μm and a fraction difference index of 0.0007. It is operated at a wavelength of 1.3μm. Find the V-number and the number of modes that the fibre will support. **(V= 4.049, Nm = 8 modes)**

Electron optics

1. An electron starts from rest and moves freely in an electric field of intensity 1500V/m. Determine the force on the electron and acceleration attained by the electron. **(2.4×10^{-16}N & 2.6×10^{14} m/s^2)**
2. A proton accelerates from rest in a uniform electric field of 500 V/m. At some time later its speed is 2.5×10^6 m/s. (Given Mass of proton: 1.67×10^{-27} kg, charge of proton 1.6×10^{-19} C)
   a. Find the acceleration of proton. **(4.8×10^{10} m/s^2)**
   b. How much time does it take to reach the above velocity? **(52 μsec)**
   c. How far it has moved during that time? **(64.9m)**
   d. What is its kinetic energy at that time? **(5.2×10^{-15} J)**
3. Electrons are accelerated through a potential difference of 200V and then projected at right angles into a magnetic field of 0.01T. Calculate the velocity of the electrons on entering the field and the radius of the path subsequently described. **($v = 8.4 \times 10^6$ m/s, r = 4.8mm)**
4. An electron enters a uniform magnetic field B = 0.23 wb/m^2 at 45° angle to B. Determine the radius and pitch of the helical path, assuming electron speed to be 3×10^7 m/s. **(R = 0.52mm, p = 3.3mm)**
5. An electron beam enters from a region of potential 75 volts into a region of potential 100 volts, making an angle of 45 degrees with the direction of electric field. Find the angle at which it emerges out of the field. **(37.67degrees)**
6. Electrons accelerated under a potential of 250V enter an electric filed at an angle of incidence of 50o
7. What transverse magnetic field acting over the entire length of the CRT must be applied to cause a deflection of 3 cm on the CRT screen that is 15 cm away from the anode and if the accelerating voltage is 2000V? **(B = 4×10^{-4} wb/m^2)**